

LEDGER · 2026-08-30 · FOOD CHEMISTRY / FLAVOR ENCAPSULATION

Potassium-Coordinated Cyclodextrin Frameworks (K-γ-CD) via Ultrasound-Assisted Crystallization

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Spray-dried maltodextrin (SD-MD)
Measured advantage	Volatile flavor payload capacity: Incumbent 10-15% vs K-γ-CD >25-30% (lower bound delta 25%/15% = 1.67x, below 2.0x threshold)
Time to first revenue	5 months
Gross margin at parity	70.6%
Startup CapEx	\$44,500
Payback from first sale	1.3 months
Capital productivity	24.27x
Regulatory risk	Moderate — qualification costs reported (\$35,000 estimated)
Assessment	

Fails the superiority threshold because comparing the 25% lower bound of K-γ-CD to the 15% upper bound of spray-dried maltodextrin yields a 1.67x improvement, which is below the mandatory 2.0x step-change bar. All capital and unit economic metrics pass.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced
- no_production_runsheet
- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-30 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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